Output Explanation

Software-Defined Radio (SDR) Communication System Using GNU Radio

BPSK modulation, noisy channel, constellation, recovered bits, error bits and BER estimate

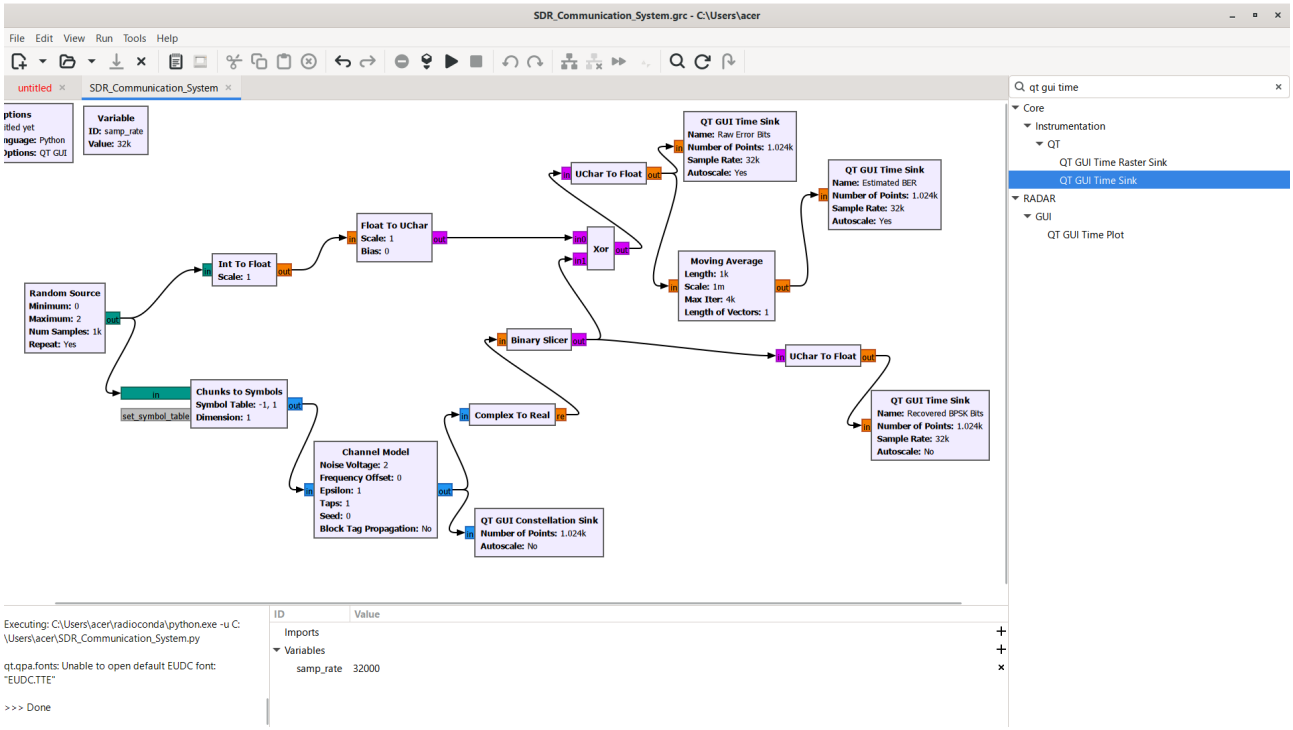


Figure 0. The GNU Radio Companion flowgraph that produced all outputs in this document.

All outputs were captured from one run with **Channel Model noise voltage = 2**, sample rate 32 kHz, and a BPSK symbol table of [-1, 1].

1. Overview of the Outputs

The flowgraph has four visual outputs. Each one observes a different point in the signal chain, so reading them in order tells the full story of what happened to the data.

#	Output	Where it is tapped	Question it answers
2	Constellation Plot	Channel Model output (complex)	How badly did noise corrupt the signal?
3	Recovered BPSK Bits	Binary Slicer output	What did the receiver decide?
4	Raw Error Bits	XOR output	Which individual bits were wrong?
5	Estimated BER	Moving Average output	What fraction of bits are wrong?

1.1 Signal chain in one paragraph

A Random Source emits 0s and 1s. Chunks to Symbols maps 0 to -1 and 1 to +1 (BPSK, real axis only). The Channel Model adds complex Gaussian noise. The receiver keeps the real part (Complex To Real) and the Binary Slicer decides 0 for negative values and 1 for non-negative values. The original bits are converted to bytes and XORed with the decisions: a 1 means mismatch. A Moving Average of 1000 error indicators with scale 0.001 gives the fraction of bad bits.

1.2 Timing used by all time plots

The sample rate is 32,000 samples/s, so one bit lasts 31.25 µs. Each time sink shows 1,024 points, which is 1024 / 32000 = 32 ms. This is why the horizontal axes run from 0 to about 32 ms.

2. Output 1: Constellation Plot

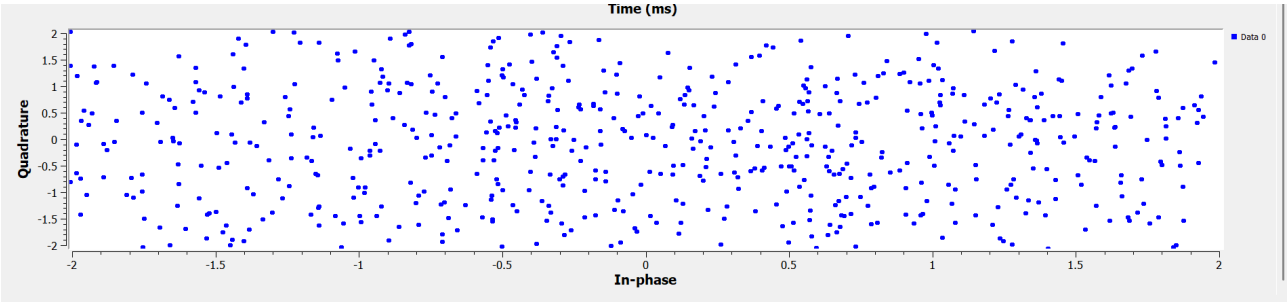


Figure 1. QT GUI Constellation Sink, noise voltage 2 (axes: In-phase vs Quadrature).

2.1 What this output is

A constellation plot draws each received complex sample as a dot: the horizontal axis is the in-phase (real) component and the vertical axis is the quadrature (imaginary) component. It is the most direct picture of signal quality, because it is taken before any bit decision is made.

2.2 What it should look like without noise

BPSK in this project uses only two symbols, -1 + 0j and +1 + 0j. A noiseless plot would therefore show just two dots on the horizontal axis, at -1 and +1.

2.3 What the screenshot shows

- The dots spread over roughly -2 to +2 on **both** axes (the plot window is limited to this range, so some points fall outside it).
- There are no two separate clusters. The points form one broad, fairly uniform cloud with no gap at 0.
- Spread on the vertical axis is as large as on the horizontal axis. The transmitted signal has zero quadrature part, so all vertical spread is pure noise. That confirms the noise is complex and affects both axes equally.

2.4 Interpretation

The signal points sit at distance 1 from the decision boundary (the vertical line at 0), while the noise spreads samples by more than that. Many samples therefore land on the wrong side of 0 and will be decoded incorrectly. This explains why the later plots show frequent errors. With a much lower noise voltage (the earlier 0.1 test) the same plot collapses into two tight clusters at -1 and +1.

2.5 Parameter link

Controlled by **Channel Model: Noise Voltage = 2**. Reducing it tightens the cloud; increasing it widens the cloud further.

3. Output 2: Recovered BPSK Bits

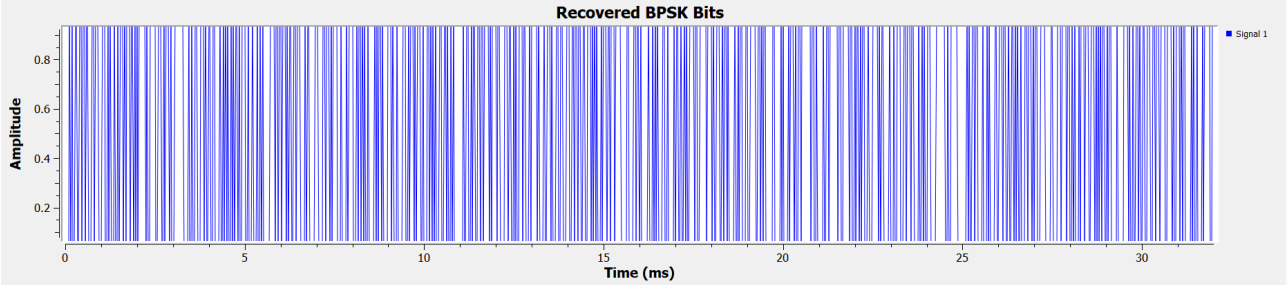


Figure 2. QT GUI Time Sink named 'Recovered BPSK Bits' (amplitude vs time in ms).

3.1 What this output is

This is the receiver's decision stream. The Complex To Real block removes the quadrature part, the Binary Slicer converts each real sample to 0 (negative) or 1 (non-negative), and UChar To Float turns the byte into a float so the time sink can plot it.

3.2 What the screenshot shows

- A dense train of vertical lines between about 0.07 and 0.93 on the y-axis. The stream only takes two values (0 and 1); the y-range is not exactly 0 to 1 because autoscale is off for this sink, so the plotted range is slightly inset.
- Bits switch quickly and irregularly, with no repeating pattern. At 32 kHz the 1,024 points cover 32 ms.
- Gaps with no transitions (for example near 3 ms and 11 ms) are runs of several identical bits, which is normal for random data.

3.3 Interpretation

Random-looking output is expected, because the transmitted data is random. This plot alone cannot say whether the bits are right or wrong; you need the original bits to compare. That comparison is what the XOR stage (next section) does. Its main value is confirming that the receiver chain produces a clean binary stream.

3.4 Parameter link

Number of Points = 1.024k, Sample Rate = 32k, Autoscale = No.

4. Output 3: Raw Error Bits

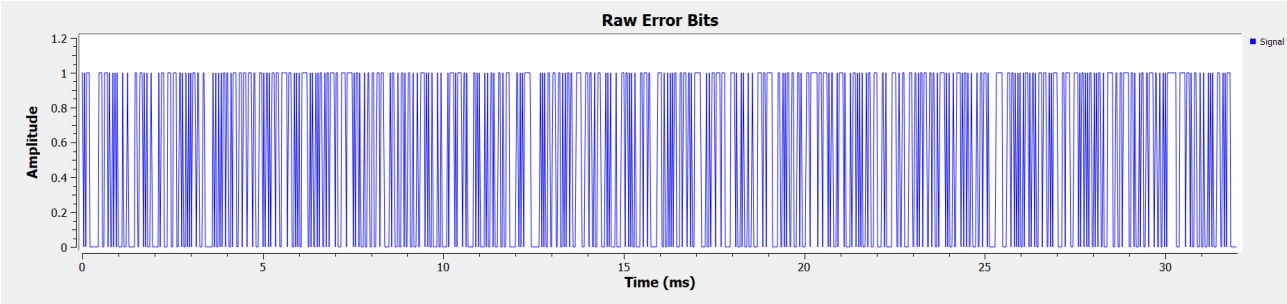


Figure 3. QT GUI Time Sink named 'Raw Error Bits' (XOR of transmitted and recovered bits).

4.1 What this output is

The XOR block compares the original bit with the recovered bit at every sample. Its truth table is simple: equal inputs give 0, different inputs give 1. The plot therefore shows a **1** wherever the receiver made a mistake, and **0** where it was correct.

4.2 What the screenshot shows

- The trace jumps between 0 and 1 constantly across the full 32 ms window.
- Pulses at 1 are very frequent. Several consecutive wide pulses mean back-to-back errors; narrow pulses are isolated errors.
- By eye, the signal spends a large share of the time at 1, roughly comparable to the time at 0.

4.3 Interpretation

A reliable link would show a mostly flat line at 0 with only rare spikes to 1. This plot is the opposite, so many bits are wrong. The next plot turns this visual impression into a number.

4.4 What 'good' would look like

With a low noise voltage such as 0.1 and aligned streams, this trace would sit at 0 almost all the time. Seeing it at 0 at zero noise is also the correct test that the transmitted and recovered streams are properly aligned (see Section 6).

5. Output 4: Estimated BER

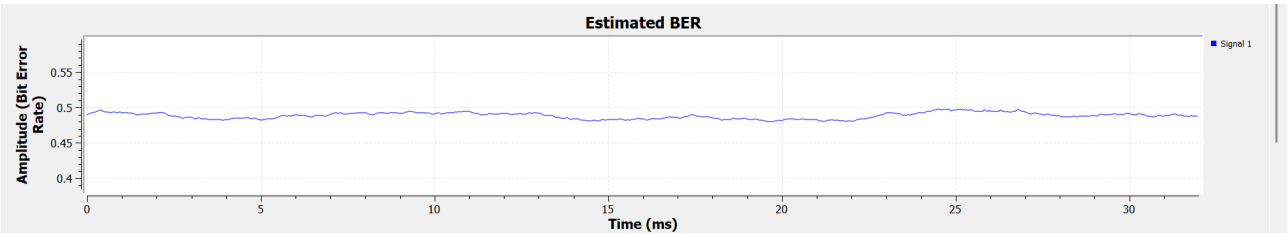


Figure 4. QT GUI Time Sink named 'Estimated BER' (y-axis: Bit Error Rate).

5.1 What this output is

The Moving Average block sums the last 1000 error indicators and multiplies by 0.001. The result is the fraction of wrong bits in the most recent 1000 bits:

BER (window) = incorrect bits in the last 1000 bits / 1000

5.2 What the screenshot shows

- A smooth curve hovering between about **0.48 and 0.50** for the entire 32 ms.
- Small slow wobbles of about ±0.01, which is statistical fluctuation of a 1000-bit window.
- The y-axis is labelled 'Bit Error Rate', and 0.5 sits near the middle of the plot.

5.3 Interpretation

A BER of 0.5 means half the bits are wrong. That is the performance of random guessing: a receiver that ignored the signal and flipped a coin would score the same. So this measurement says the link is essentially not working at noise voltage 2, or that the measurement is not comparing the correct pairs of bits.

5.4 Why 0.5 is a red flag, not just 'lots of noise'

For BPSK in Gaussian noise the expected BER is $Q(1/\sigma)$, where σ is the noise standard deviation per real component. Assuming GNU Radio's complex noise of voltage 2 has total power 4, so $\sigma \approx 1.41$, the theoretical BER is $Q(0.71) \approx$ **0.24**, about half of what is measured. (This assumption about GNU Radio's noise scaling should be double-checked for your version; even with a different scaling, BER climbing all the way to 0.5 at an amplitude-2 noise is unlikely for a correctly aligned BPSK link.)

Quantity	Value
Measured BER (this screenshot)	≈ 0.49
Theoretical BER, $\sigma \approx 1.41$ (assumed scaling)	≈ 0.24
BER of random guessing	0.50

6. Putting the Outputs Together

6.1 What the four outputs say as a group

- **Constellation:** noise is large compared with the signal, so a high BER is expected.
- **Recovered bits:** the receiver chain works mechanically and produces a clean binary stream.
- **Raw error bits:** errors are frequent and spread over the whole window.
- **Estimated BER:** the error rate is about 0.49, consistent with the error plot, but higher than the approximately 0.24 expected from theory.

6.2 Most likely cause of the extra errors: bit misalignment

The BER measurement is only valid if the bit compared at the XOR is the *same* bit that was transmitted. In this flowgraph the reference bits go directly from the Random Source to the XOR, while the transmitted symbols pass through the Channel Model first. The Channel Model contains a fractional resampling/filtering stage that delays the signal by a few samples. Even a one-sample shift makes the two streams unrelated, so the XOR output is 1 about half the time whatever the noise. That matches the measured value of about 0.5.

This is a hypothesis based on the plots and the block settings, not a confirmed diagnosis. It is easy to test.

6.3 How to verify and fix

- Set Channel Model noise voltage to **0** and run. If the Estimated BER is 0, the streams are aligned and the 0.49 at noise 2 would then be a genuine noise result (which would be surprising versus 0.24, so re-check the noise scaling).
- If the Estimated BER at zero noise is still about 0.5, the streams are misaligned. Insert a **Delay** block on the reference branch (before the XOR) and try small values (for example 1 to 8 samples) until BER reads 0 at zero noise.
- Keep that delay, set noise voltage back to 2, and compare with the theoretical value of about 0.24.
- Then sweep the noise voltage (for example 0.2, 0.5, 1, 1.5, 2) and record BER for each to build a BER-vs-noise table.

6.4 What can be claimed from the current results

- The simulation runs and all four displays work.
- Increasing noise visibly degrades the constellation and produces many bit errors.
- The numeric BER of about 0.49 should **not** be quoted as the performance of BPSK at noise voltage 2 until alignment has been verified.